LECTURE 10

Classification

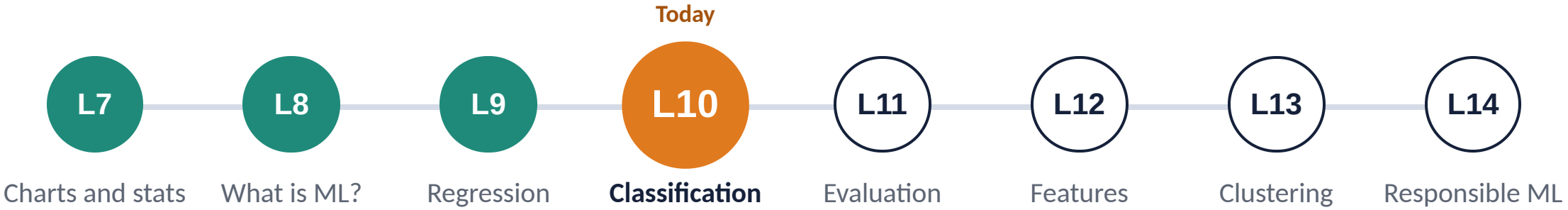
k-nearest neighbours and decision trees

Today's key line of code

```
KNeighborsClassifier(n_neighbors=3)
DecisionTreeClassifier(max_depth=2)
```

Introduction to Data Science and Machine Learning · Md Mahbub E Noor · University of Barishal

The road ahead: machine learning



Today you will

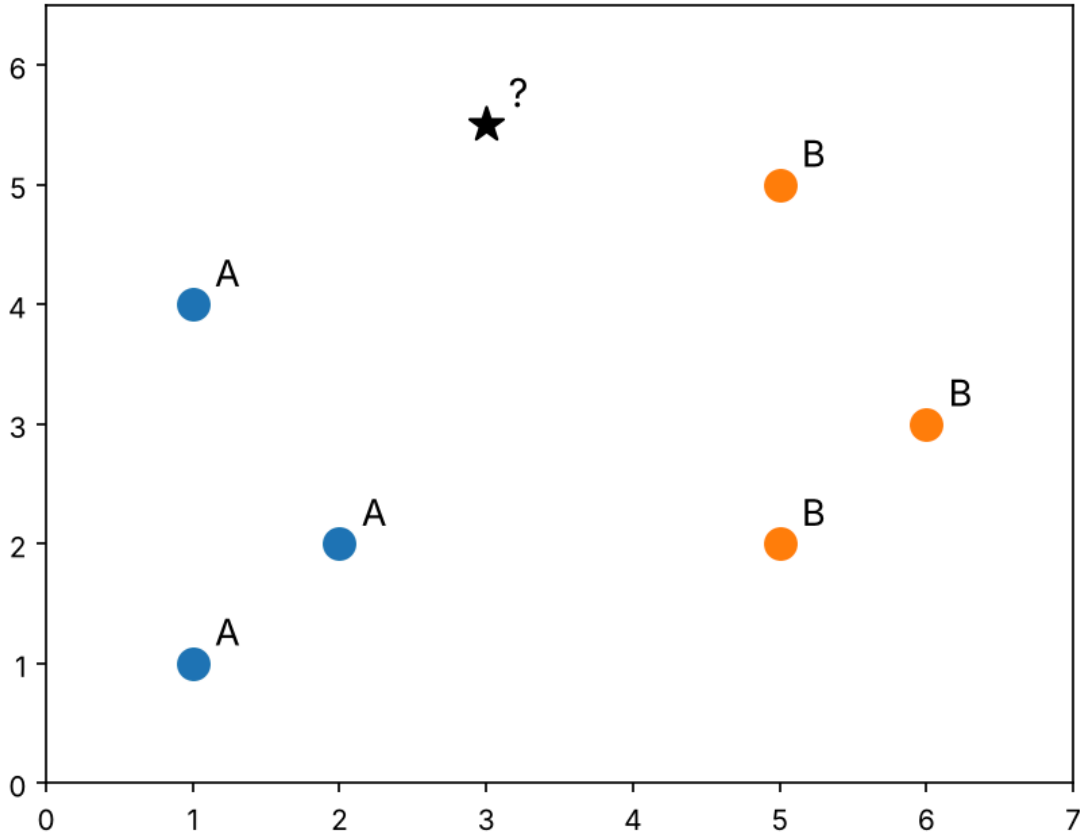
1. Classify a new point by hand with k-nearest neighbours
2. See how the choice of k changes the answer
3. Read the rules a decision tree learns
4. Follow a tree by hand to classify a flower
5. Compare the two models and choose sensibly

Six labelled points and one new point

Class A in blue, class B in orange. Which class should the new point (?) get?

Code

```
import matplotlib.pyplot as plt
pts = [(1, 1, "A"), (2, 2, "A"), (1, 4, "A"),
       (5, 5, "B"), (6, 3, "B"), (5, 2, "B")]
for x, y, c in pts:
    col = "tab:blue" if c == "A" else "tab:orange"
    plt.scatter(x, y, s=120, color=col)
    plt.annotate(c, (x + 0.15, y + 0.15))
plt.scatter(3, 5.5, s=160, marker="*", color="k")
plt.annotate("?", (3.15, 5.65), fontsize=14)
plt.xlim(0, 7); plt.ylim(0, 6.5)
plt.show()
```



By hand: distances to the new point (3, 5.5)

Distance = $\sqrt{((x - 3)^2 + (y - 5.5)^2)}$. Sort by distance, then let the k nearest vote.

Code

```
# k = 1: nearest is B (2.06)          -> B
# k = 3: B (2.06), A (2.5), A (3.64)
#        A has 2 votes                -> A
# Same data, different k, different answer!
```

rank	point	class	distance
1	(5, 5)	B	2.06
2	(1, 4)	A	2.50
3	(2, 2)	A	3.64
4	(6, 3)	B	3.91
5	(5, 2)	B	4.03
6	(1, 1)	A	4.92

Next row: arrow key or the ▶ button.
Predict it first!

Watch it: k-nearest neighbours voting

WATCH FOR

- A circle grows from the new point
- It stops when it holds k neighbours
- The neighbours vote; the majority wins
- With k = 1 and k = 3 the answer differs



Silent video: press play, pause any time.

k-nearest neighbours in scikit-learn

- `n_neighbors` is k
- Small k: follows every single point (can overfit)
- Large k: smoother, but may ignore real detail (here k = 50 drops to 0.867)
- Check several k on the test set… for now

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
X, y = load_iris(return_X_y=True)
X_tr, X_te, y_tr, y_te = train_test_split(
    X, y, test_size=0.3, random_state=1, stratify=y)
for k in [1, 3, 5, 15, 50]:
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_tr, y_tr)
    print(k, round(knn.score(X_te, y_te), 3))
```

Output

```
1 0.978
3 0.978
5 0.978
15 0.978
50 0.867
```

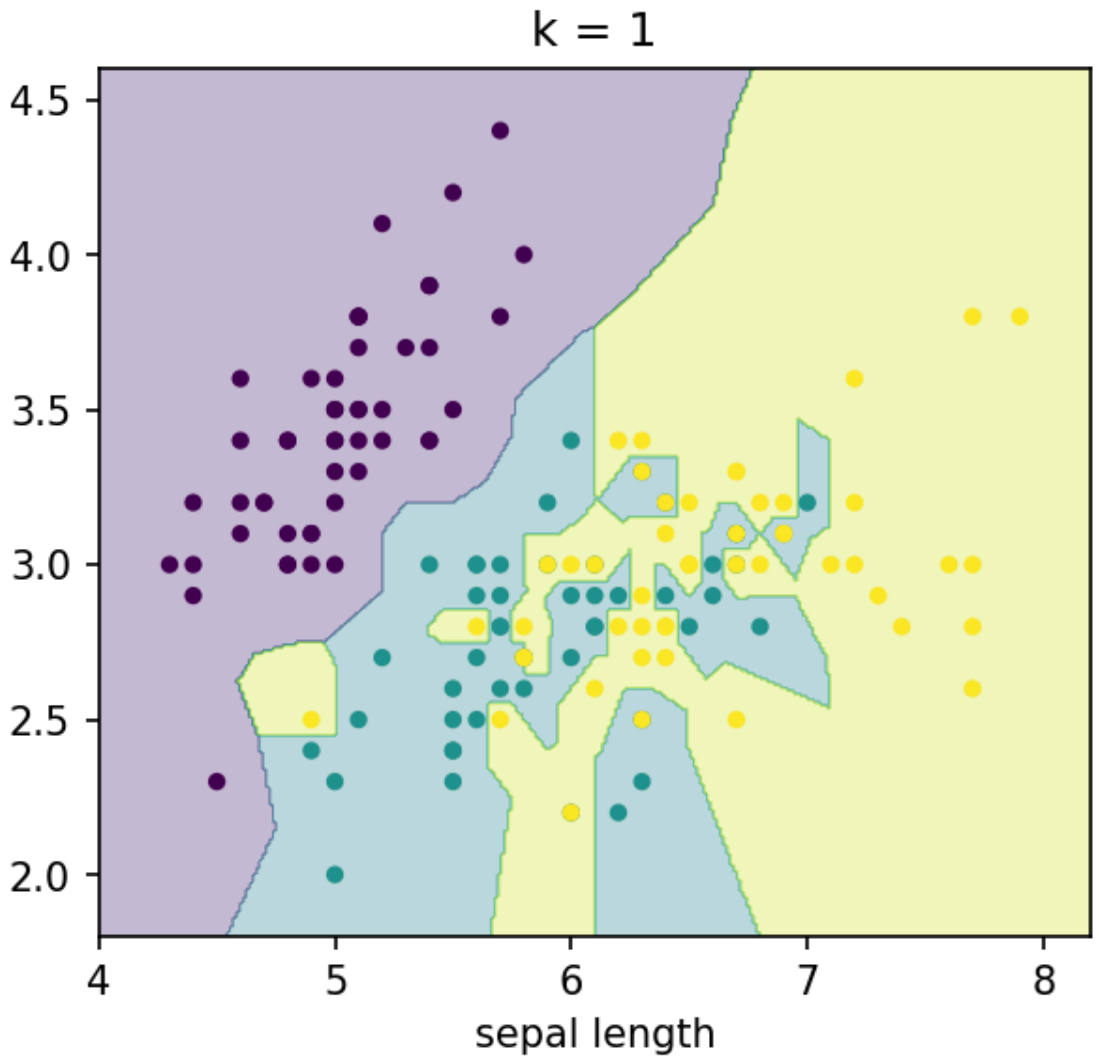
Decision boundaries: small k vs large k

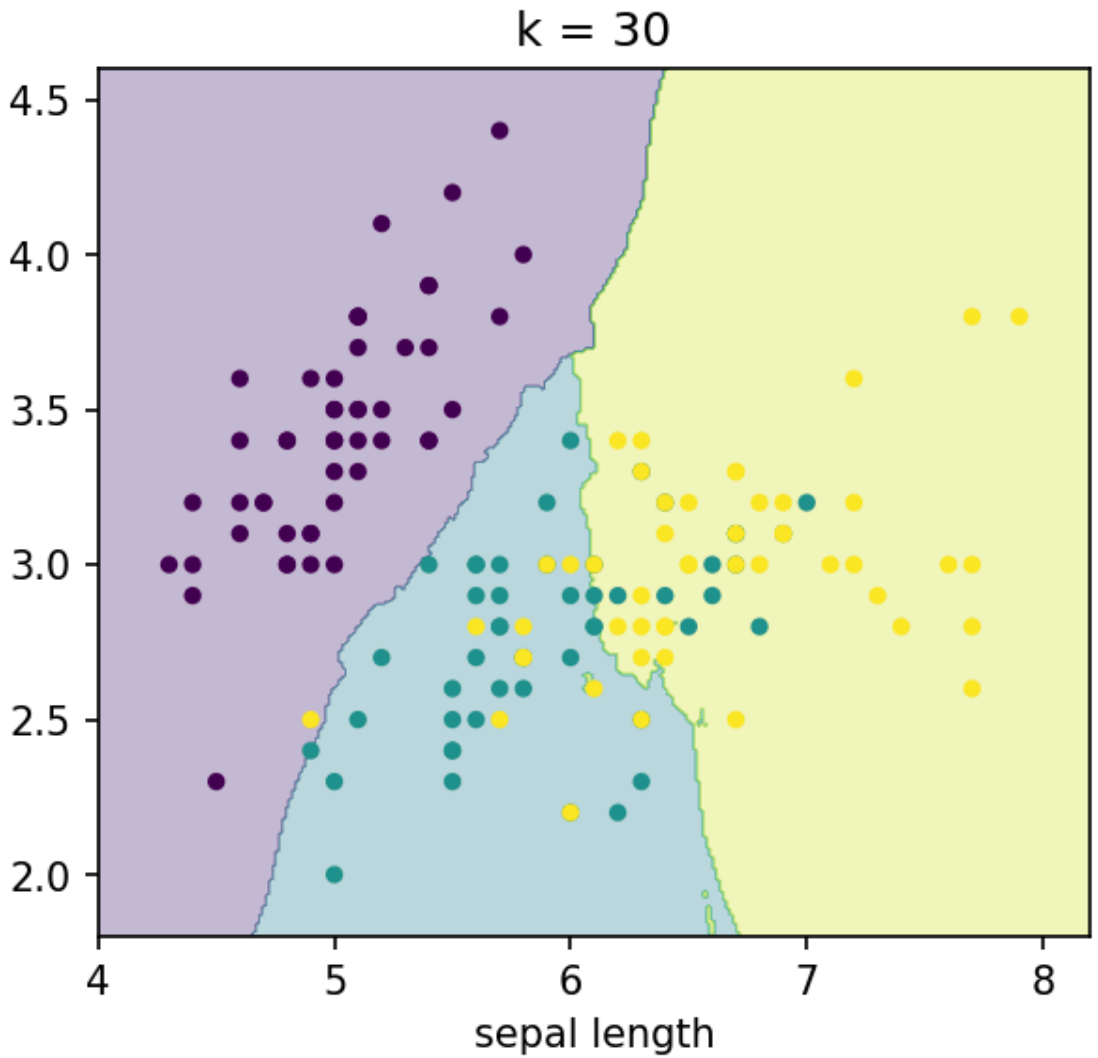
Two sepal measurements, where the species overlap. Colours show the predicted class at every point: k = 1 draws jagged islands, k = 30 smooth regions.

Code

```
import numpy as np
X2 = X[:, 0:2]
xx, yy = np.meshgrid(np.linspace(4, 8.2, 300),
                     np.linspace(1.8, 4.6, 300))
grid = np.c_[xx.ravel(), yy.ravel()]
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
for ax, k in zip(axes, [1, 30]):
    m = KNeighborsClassifier(n_neighbors=k).fit(X2, y)
    zz = m.predict(grid).reshape(xx.shape)
    ax.contourf(xx, yy, zz, alpha=0.3, cmap="viridis")
    ax.scatter(X2[:, 0], X2[:, 1], c=y, s=12, cmap="viridis")
    ax.set_title(f"k = {k}")
    ax.set_xlabel("sepal length")
plt.show()
```

Decision boundaries: small k vs large k: the result





A decision tree plays twenty questions

1

Ask a question

Is petal width at most 0.75 cm? Each question splits the data in two.

2

Pick the best question

The tree chooses the question that makes each side as pure as possible.

3

Repeat

Keep asking until the groups are pure or a depth limit is reached. Leaves give the answer.

Train a tree and read its rules

`max_depth=2`: at most two questions per flower · `export_text` prints the learned rules · class 0 = setosa, 1 = versicolor, 2 = virginica · Nobody wrote these rules: the data did

Code

```
from sklearn.tree import DecisionTreeClassifier, export_text
names = ["sepal len", "sepal wid", "petal len", "petal wid"]
tree = DecisionTreeClassifier(max_depth=2, random_state=0)
tree.fit(X_tr, y_tr)
print(export_text(tree, feature_names=names))
print(round(tree.score(X_te, y_te), 3))
```

Output

```
|--- petal wid <= 0.75
|   |--- class: 0
|--- petal wid >  0.75
|   |--- petal len <= 4.75
|   |   |--- class: 1
|   |--- petal len >  4.75
|   |   |--- class: 2

0.956
```

By hand: follow the tree for three flowers

Use the two rules from the previous slide. Then check: the model's own predictions are in the last column.

Code

```
if petal_wid <= 0.75:
    -> setosa (0)
elif petal_len <= 4.75:
    -> versicolor (1)
else:
    -> virginica (2)
```

petal len	petal wid	path	model says
1.5	0.2	wid <= 0.75	**0**
4.7	1.2	wid > 0.75, len <= 4.75	**1**
5.8	2.2	wid > 0.75, len > 4.75	**2**

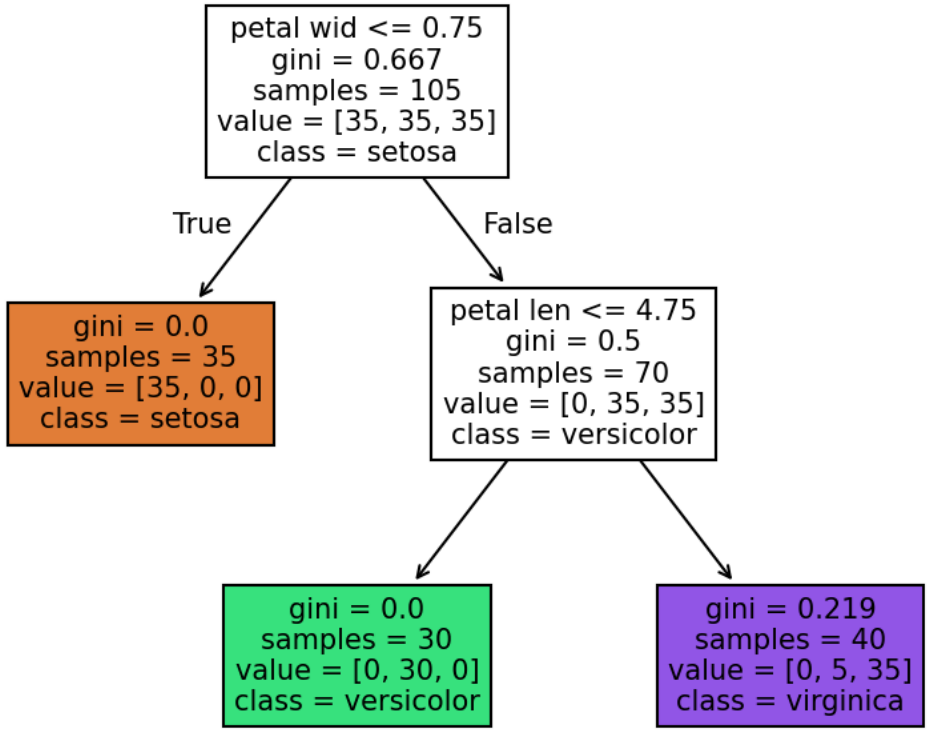
Next row: arrow key or the ▶ button. Predict it first!

Draw the tree

Each box shows the question, how many flowers reach it, and the class counts. Leaves at the bottom give the answer.

Code

```
from sklearn.tree import plot_tree
classes = ["setosa", "versicolor", "virginica"]
plt.figure(figsize=(7, 5.5))
plot_tree(tree, feature_names=names,
          class_names=classes,
          filled=True, fontsize=10)
plt.show()
```



Watch it: a tree splitting the data

WATCH FOR

- First question: a horizontal cut at petal width 0.75
- Setosa is separated completely
- Second question: a vertical cut at petal length 4.75
- Each region gets the majority class



Silent video: press play, pause any time.

k-nearest neighbours or decision tree?

k-nearest neighbours

- No real training: remembers all data
- Slow to predict on big data
- Needs features on similar scales
- Hard to explain a single answer

Decision tree

- Learns explicit if/else rules
- Fast to predict
- Scale of features does not matter
- Deep trees overfit easily (Lecture 11)

Your turn 10 minutes

1. Train DecisionTreeClassifier with max_depth 1, 2, 3, 10 on iris
2. Print training and test accuracy for each
3. Which depth would you choose, and why?
4. Bonus: print export_text for depth 3

Code

```
for d in [1, 2, 3, 10]:
    t = DecisionTreeClassifier(max_depth=d, random_state=0)
    t.fit(X_tr, y_tr)
    print(d, t.score(X_tr, y_tr), t.score(X_te, y_te))
```

Common mistakes with classifiers

Read the last line of an error first. It names the problem.

What you typed	What Python says	What it means
`KNeighborsClassifier().predict([[1, 2, 3, 4]])`	`NotFittedError: This KNeighborsClassifier instance is not fitted yet. Call 'fit' with appropriate arguments before using this estimator.`	Not fitted yet. Call fit before predict.

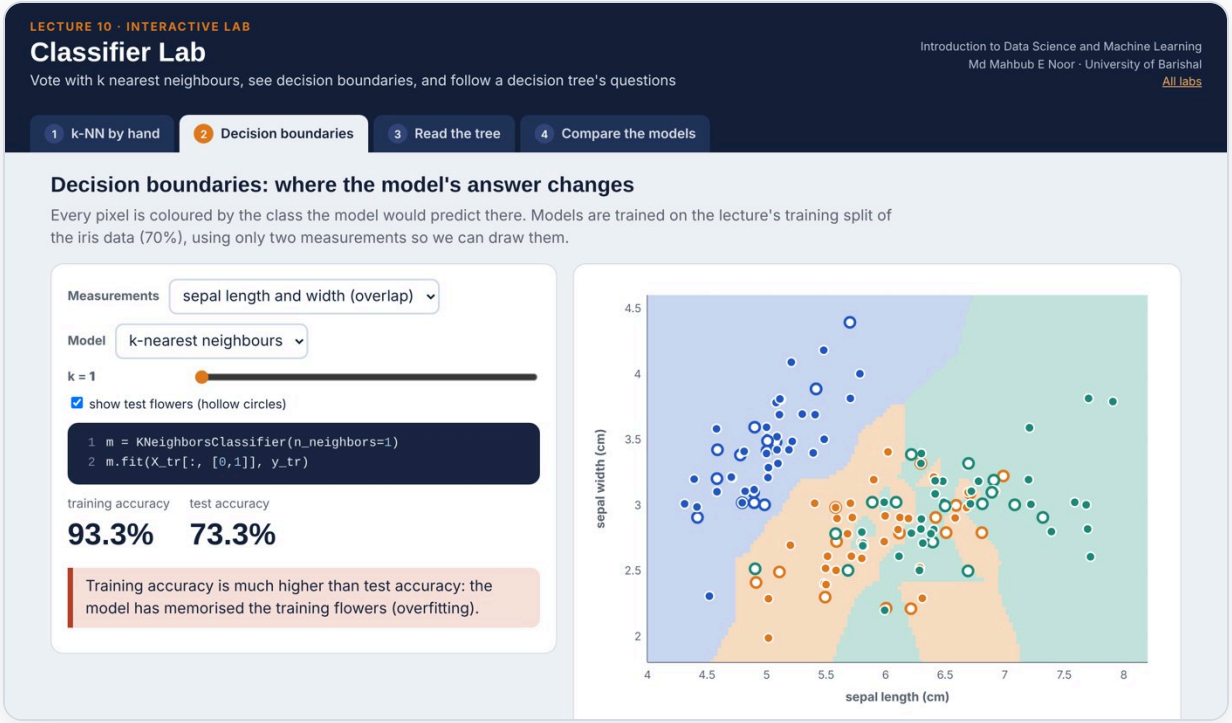
`knn = KNeighborsClassifier(n_neighbors=200)`
`knn.fit(X_tr, y_tr)`
`knn.predict(X_te)`
	`ValueError: Expected n_neighbors <= n_samples_fit, but n_neighbors = 200, n_samples_fit = 105, n_samples = 45`	k cannot be larger than the number of training rows (105).

`tree.predict([[1.5, 0.2]])`	`ValueError: X has 2 features, but DecisionTreeClassifier is expecting 4 features as input.`	The model was trained on 4 features; give it 4 values.

Interactive lab: Classifier Lab

In the lab you can

1. **k-NN by hand**
   Click to move the new point; the k nearest neighbours vote.
2. **Decision boundaries**
   k-NN and trees on two iris measurements, train and test.
3. **Read the tree**
   Slide a flower's measurements and follow its path.
4. **Compare the models**
   Accuracy against k and depth, and why units matter for k-NN.



Open the lab ▶

Opens in a new tab and works offline. All labs: labs/START_HERE.html

Today's toolkit

`KNeighborsClassifier(n_neighbors=k)`	Vote of the k nearest
`distance`	$\sqrt{(\Delta x^2 + \Delta y^2)}$
`DecisionTreeClassifier(max_depth=d)`	Learned if/else rules
`export_text(tree)`	Print the rules

`plot_tree(tree)`	Draw the tree
`decision boundary`	Where predictions change
`k, max_depth`	Hyperparameters we choose

Homework 10

Due before Practice 1 · [date]

1 By hand

For the six points and new point (4, 4), compute all distances and predict with k = 1, 3 and 5.

2 Choose k

On the wine dataset, print test accuracy for k = 1, 3, 5, 9, 15. Which k is best?

3 Tree rules

Train a depth-2 tree on wine and write its rules as if/elif statements in a text cell.

4 Boundary

Adapt the boundary plot for a decision tree of depth 2 and depth 10. Describe the difference.

5 Explain

In 3 sentences: when would you prefer a decision tree to k-nearest neighbours?

Submit on [LMS / Google Classroom]: your notebook file HW10_<your ID>.ipynb, or a Colab link set to “Anyone with the link can view”.

Recap: Lecture 10 in 90 seconds

RECAP

Lecture 10 in 90 seconds

A short recap of today's lecture. Pause any time.

Watch it again at home to revise.